Test Series No.

1.14

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 24/03/2026

ANSWER KEY

1.	(1)	46.	(2)	91.	(2)	136.	(3)
2.	(4)	47.	(1)	92.	(2)	137.	(2)
3.	(4)	48.	(2)	93.	(1)	138.	(3)
4.	(2)	49.	(3)	94.	(4)	139.	(2)
5.	(3)	50.	(4)	95.	(3)	140.	(1)
6.	(4)	51.	(2)	96.	(2)	141.	(3)
7.	(1)	52.	(1)	97.	(1)	142.	(1)
8.	(3)	53.	(3)	98.	(2)	143.	(3)
9.	(4)	54.	(2)	99.	(2)	144.	(3)
10.	(1)	55.	(1)	100.	(3)	145.	(3)
11.	(2)	56.	(3)	101.	(1)	146.	(2)
12.	(4)	57.	(4)	102.	(2)	147.	(4)
13.	(4)	58.	(3)	103.	(3)	148.	(2)
14.	(4)	59.	(3)	104.	(1)	149.	(2)
15.	(1)	60.	(2)	105.	(4)	150.	(1)
16.	(2)	61.	(3)	106.	(4)	151.	(3)
17.	(3)	62.	(1)	107.	(3)	152.	(2)
18.	(2)	63.	(3)	108.	(3)	153.	(1)
19.	(3)	64.	(2)	109.	(3)	154.	(3)
20.	(2)	65.	(4)	110.	(2)	155.	(2)
21.	(3)	66.	(4)	111.	(2)	156.	(2)
22.	(1)	67.	(2)	112.	(3)	157.	(3)
23.	(1)	68.	(4)	113.	(2)	158.	(1)
24.	(3)	69.	(1)	114.	(1)	159.	(3)
25.	(1)	70.	(4)	115.	(2)	160.	(2)
26.	(1)	71.	(3)	116.	(1)	161.	(3)
27.	(2)	72.	(2)	117.	(1)	162.	(4)
28.	(4)	73.	(2)	118.	(3)	163.	(2)
29.	(1)	74.	(1)	119.	(2)	164.	(4)
30.	(1)	75.	(3)	120.	(1)	165.	(2)
31.	(1)	76.	(1)	121.	(1)	166.	(3)
32.	(4)	77.	(3)	122.	(2)	167.	(4)
33.	(2)	78.	(3)	123.	(3)	168.	(3)
34.	(3)	79.	(1)	124.	(2)	169.	(2)
35.	(1)	80.	(1)	125.	(3)	170.	(3)
36.	(1)	81.	(4)	126.	(3)	171.	(1)
37.	(2)	82.	(2)	127.	(4)	172.	(2)
38.	(1)	83.	(1)	128.	(3)	173.	(1)
39.	(3)	84.	(2)	129.	(4)	174.	(4)
40.	(2)	85.	(3)	130.	(4)	175.	(3)
41.	(4)	86.	(4)	131.	(1)	176.	(3)
42.	(3)	87.	(2)	132.	(1)	177.	(3)
43.	(1)	88.	(1)	133.	(2)	178.	(3)
44.	(1)	89.	(3)	134.	(2)	179.	(1)
45.	(1)	90.	(3)	135.	(3)	180.	(2)

[Contd...

1. (1)

Angular Momentum conserved ⇒ Iω = constant.

Volume increases by 3%

⇒ Radius increases by 1%

⇒ $I \propto R^2$ ⇒ I increases by 2%.

Hence ω decreases by 2%.

(NCERT Ed. 2025-26, 11th, Page No. 122)

2. (4)

Balancing torque about the hinge,

$$mg\frac{L}{2}\sin 60^\circ = Mg\frac{L}{2}\sin 30^\circ$$

$$\Rightarrow \frac{M}{m} = \frac{\sin 60^\circ}{\sin 30^\circ} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$$

(NCERT Ed. 2025-26, 11th, Page No. 111)

3. (4)

Given r_0 = (2,3) m and r = (8,–5) m

⇒ R = (6,–8), |R| = 10 m.

$$E = \frac{kq}{R^2} = \frac{(9\times10^9)(50\times10^{-6})}{100} = 4500 \text{ V/m}.$$

(NCERT Ed. 2025-26, 12th, Page No. 14)

4. (2)

Using Gauss law inside sphere:

$$E\left(4\pi r^2\right) = \frac{\rho\left(\frac{4}{3}\pi r^3\right)}{\varepsilon_0}$$

$$\Rightarrow E = \frac{\rho r}{3\varepsilon_0}.$$

(NCERT Ed. 2025-26, 12th, Page No. 36)

5. (3)

$$B(x) = \mu_0 \frac{NI\,R^2}{2\left(R^2+x^2\right)^{\frac{3}{2}}}.$$

Given $B(x) = \frac{1}{2}B(0)$

$$\Rightarrow \frac{B(0)}{B(x)} = 2 = \left(1+\frac{x^2}{R^2}\right)^{\frac{3}{2}}$$

$$\Rightarrow \left(1+\frac{x^2}{R^2}\right) = 2^{\frac{3}{2}} = 1.587$$

$$\Rightarrow x = \sqrt{0.587}\,R = 0.766 \text{ R}$$

(NCERT Ed. 2025-26, 12th, Page No. 116)

6. (4)

Carnot $\eta = 1 - \frac{T_2}{T_1}$ (in Kelvin).

$\eta_1 = \frac{200}{473}$, $\eta_2 = \frac{200}{273}$

⇒ ratio = $\frac{273}{473}$.

(NCERT Ed. 2025-26, 11th, Page No. 238)

7. (1)

Collision rate $\propto n\sqrt{T} \propto \frac{\sqrt{T}}{V}$

$$\Rightarrow V \propto \sqrt{T}$$

$$PV = nRT \Rightarrow P \propto \sqrt{T} \Rightarrow P \propto V$$

(NCERT Ed. 2025-26, 11th, Page No. 249)

8. (3)

V || PQ, so no emf is induced in PQ.

$$L_{eff} = 2L\sin\left(\frac{\theta}{2}\right)$$

Induced emf between ends X and Y.

$$e = 2BLV\sin\left(\frac{\theta}{2}\right)$$

(NCERT Ed. 2025-26, 12th, Page No. 162)

9. (4)

$$\frac{L}{(RCV)} \Rightarrow \frac{(\Omega.\text{s})}{(\Omega.\text{C})} = \frac{\text{s}}{\text{C}} = \frac{1}{A}.$$

(NCERT Ed. 2025-26, 12th, Page No. 168)

10. (1)

Equal stress ⇒ $T_A:T_B = \frac{A_{brass}}{A_{steel}} = 2:1$.

Torque balance about C,

$$T_A x = T_B(2-x)$$

$$\Rightarrow 2x = 2 - x$$

$$\Rightarrow x = 0.666 \text{ m}$$

(NCERT Ed. 2025-26, 11th, Page No. 168)

11. (2)

$$v = \sqrt{108-9x^2}\ ;\ \frac{dv}{dx} = \frac{1}{2\sqrt{108-9x^2}}(-18x)$$

$$a = v\frac{dv}{dx} = -9x = -\omega^2 x$$

so motion is non uniformly accelerated.

at x = 3 m, a = –27 m/s² ⇒|a| = 27 m/s²

at $x = A$, $v = 0 \Rightarrow A = 2\sqrt{3}$ m.

(NCERT Ed. 2025-26, 11th, Page No. 18)

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12. (4)

Tension $\propto$ y $\Rightarrow v=\sqrt{(gy)}$.

Time taken $\approx =2\sqrt{\frac{L}{g}}=2$ s

(NCERT Ed. 2025-26, 11th, Page No. 285)

13. (4)

In YDSE,

$I = I_{max}\cos^2\left(\frac{\Phi}{2}\right)$

$\Rightarrow \cos^2\left(\frac{\Phi}{2}\right)=\frac{1}{2} \Rightarrow \cos\left(\frac{\Phi}{2}\right)=\frac{1}{\sqrt{2}}=\cos\left(\frac{\pi}{4}\right)$

$\Rightarrow \Phi=\frac{\pi}{2}$

$\Rightarrow \left(\frac{2\pi}{\lambda}\right)\Delta z=\frac{\pi}{2} \Rightarrow \Delta z=\frac{\lambda}{4}$

$\Rightarrow \frac{yd}{D}=\frac{\lambda}{4} \Rightarrow y=\frac{\lambda D}{4d}$

(NCERT Ed. 2025-26, 12th, Page No. 264)

14. (4)

For a mirror,

$m=\frac{f}{f-u}$

Case I: $m_1=\frac{f}{f+25}$

Case II: $m_2=\frac{f}{f+40}$

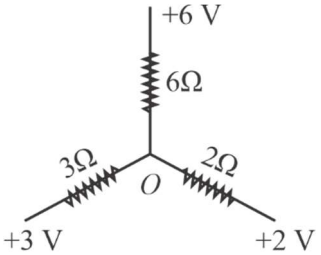
$\frac{m_1}{m_2}=\frac{f+40}{f+25}=4$

$\Rightarrow$4f + 100 = f + 40

$\Rightarrow$f = –20 cm (Concave mirror).

(NCERT Ed. 2025-26, 12th, Page No. 225)

15. (1)



Applying KCL at O,

$\frac{6-V_0}{6}+\frac{3-V_0}{3}+\frac{2-V_0}{2}=0$

$\Rightarrow \frac{6-V_0+2(3-V_0)+3(2-V_0)}{6}=0$

$\Rightarrow 6\ V_0 = 18 \Rightarrow V_0 = 3$ V

(NCERT Ed. 2025-26, 12th, Page No. 97)

16. (2)

In displacement method,

$f=\frac{(D^2-x^2)}{4D}=\frac{40}{3}$ cm.

(NCERT Ed. 2025-26, 12th, Page No. 225)

17. (3)

For node at x=0, need opposite travelling wave:

$y_2 = -a\cos(kx + \omega t)$.

(NCERT Ed. 2025-26, 11th, Page No. 290)

18. (2)

$\lambda=\frac{h}{\sqrt{(2mE)}}\approx 1.43\ \mathring{A}$.

(NCERT Ed. 2025-26, 12th, Page No. 285)

19. (3)

Energy released = 28 − 4.4 = 23.6 MeV.

(NCERT Ed. 2025-26, 12th, Page No. 315)

20. (2)

$\frac{\mu}{L}=\frac{e}{2}m$.

(NCERT Ed. 2025-26, 12th, Page No. 124)

21. (3)

$\sin\theta_c=\frac{3}{4} \Rightarrow \tan\theta_c=\frac{3}{\sqrt{7}}$.

Radius = $12\times\frac{3}{\sqrt{7}}$.

(NCERT Ed. 2025-26, 12th, Page No. 230)

22. (1)

Both true; conservation of angular momentum explains it.

(NCERT Ed. 2025-26, 11th, Page No. 128)

[Contd...

23. (1)

Total pressure at the bottom,

$= P_{atm} + \rho g(2h) + 2\rho g(2h) = P_{atm} + 6\rho gh$

Applying Bernoulli's equation,

$P_{atm} + 6\rho gh = P_{atm} + \frac{1}{2}(2\rho)v^2$

$\Rightarrow v = \sqrt{6gh}$

(NCERT Ed. 2025-26, 11th, Page No. 189)

24. (3)

As soap bubble has two surfaces,

work done = $8\pi Tr^2 \approx 75.36$ μJ.

(NCERT Ed. 2025-26, 11th, Page No. 196)

25. (1)

k = 10 N/m, $\omega = \frac{\pi}{4}$ rad s^{-1}.

$m = \frac{160}{\pi^2}$, $KE_{max} = 8\times10^{-3}$ J

(NCERT Ed. 2025-26, 11th, Page No. 267)

26. (1)

Induced charge $-q$ on the inner surface is non–uniformly distributed, but equidistant from the centre O. Charge $+q$ on the outer surface is uniformly distributed.

So, potential at the centre,

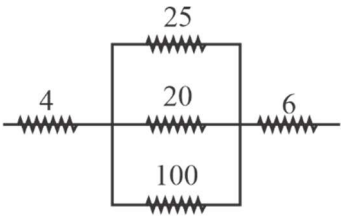
$V_0 = \frac{q}{4\pi\epsilon_0 r} - \frac{q}{4\pi\epsilon_0 R} + \frac{q}{4\pi\epsilon_0(2R)}$

$= \frac{q}{4\pi\epsilon_0}\left[\frac{1}{r} - \frac{1}{2R}\right]$

(NCERT Ed. 2025-26, 12th, Page No. 38)

27. (2)

On redrawing the circuit,



Equivalent resistance of the circuit,

$R_{eq} = 4 + 10 + 6 = 20\Omega$

V = 5 x 20 = 100 V.

(NCERT Ed. 2025-26, 12th, Page No. 101)

28. (4)

i = e = 45°, A = 60°.

δ = i + e − A = 30°.

(NCERT Ed. 2025-26, 12th, Page No. 240)

29. (1)

In RC ac circuit,

p.d. across R and current are in same phase.

p.d. across R is ahead in phase with p.d. across capacitor.

(NCERT Ed. 2025-26, 12th, Page No. 181)

30. (1)

$E = \sqrt{V_R^2 + (V_C - V_L)^2} = \sqrt{80^2 + 60^2}$

⇒E = 100 V

(NCERT Ed. 2025-26, 12th, Page No. 191)

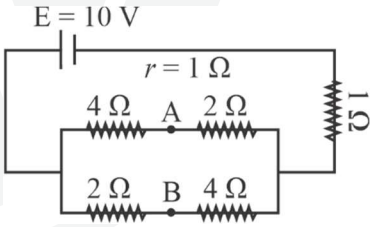
31. (1)

When current in the circuit decreases, induced emf opposes the decay of current.

Both statements (I) and (II) are correct.

(NCERT Ed. 2025-26, 12th, Page No. 168)

32. (4)



Current drawn from the battery,

$I = \frac{10}{3+1+1} = 2$ A

Now, $V_A + 4\times1 - 2\times1 = V_B$

$\Rightarrow V_A - V_B = -2$ V

(NCERT Ed. 2025-26, 12th, Page No. 101)

33. (2)

Case I: Both diodes are forward biased. P.d. across Ge diode is 0.3V, so $V_o = 11.7$ V.

Case II: Ge is reversed biased and Si diode is forward biased. P.d. across it = 0.7V, so $V_o = 11.3$ V

so, output Vo changes by 0.4 V

(NCERT Ed. 2025-26, 12th, Page No. 326)

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34. (3)

$$v_e = \sqrt{\frac{2GM}{R}}$$

$$v_{e'} = \sqrt{\frac{2G(4M)}{R/2}} = \sqrt{\frac{16GM}{R}}$$

$$\Rightarrow \frac{v_{e'}}{v_e} = \sqrt{\frac{16}{2}} = 2\sqrt{2}$$

$$\Rightarrow v_{e'} = 2\sqrt{2}v_e$$

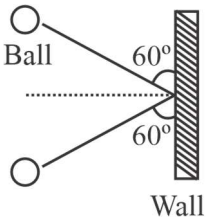
(NCERT Ed. 2025-26, 11th, Page No. 136)

35. (1)

As effective mass of electron is lesser, $\mu_e > \mu_h$.

(NCERT Ed. 2025-26, 12th, Page No. 327)

36. (1)



Angle with the normal,

$\theta = (90^\circ - 60^\circ) = 30^\circ$

$|\Delta\vec{P}| = 2mv\cos\theta$

$= 2 \times 3 \times 100\frac{\sqrt{3}}{2}$

$= 300\sqrt{3}$

$F = \frac{|\Delta\vec{P}|}{\Delta t} = \frac{300\sqrt{3}}{0.2} = 1500\sqrt{3}$ N

(NCERT Ed. 2025-26, 11th, Page No. 58)

37. (2)

$N = m(g - a) = 4$ N.

(NCERT Ed. 2025-26, 11th, Page No. 52)

38. (1)

$\tan 30^\circ = \frac{1}{\sqrt{3}} \approx 0.6$

$\mu = 0.8$

As $\mu > 0.6$, static friction is sufficient.

so, T = 0

(NCERT Ed. 2025-26, 11th, Page No. 60)

39. (3)

Power = Fv = K ⇒ Work = Kt.

(NCERT Ed. 2025-26, 11th, Page No. 75)

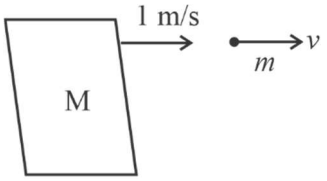
40. (2)

From energy conservation,

$$\frac{1}{2}kx^2 = mv^2 \Rightarrow x = \sqrt{\left(\frac{2mv^2}{k}\right)}.$$

(NCERT Ed. 2025-26, 11th, Page No. 76)

41. (4)



(After collision)

In elastic collision,

Velocity of separation = Velocity of approach

$v - 1 = 1 - (-4)$

$\Rightarrow v = 6$ m/s

(NCERT Ed. 2025-26, 11th, Page No. 84)

42. (3)

Area = 15.6, As breadth is having least number of significant figures (2 significant figures), so after rounding off, area = 16 cm².

(NCERT Ed. 2025-26, 11th, Page No. 5)

43. (1)

Maximum relative error in Z,

$$\frac{\Delta Z}{Z} = 2\frac{\Delta A}{A} + \frac{\Delta B}{B} + 3\frac{\Delta C}{C}.$$

(NCERT Ed. 2025-26, 11th, Page No. 7)

44. (1)

Very close to the surface of earth, orbital speed of a satellite, $v = \sqrt{GR}$.

(NCERT Ed. 2025-26, 11th, Page No. 137)

45. (1)

$Y = \overline{A} \cdot B + A \cdot \overline{B}$

For A=0, B=0; Y=0

A=0, B=1; Y=1

A=1, B=0; Y=1

A=1, B=1; Y=0

(NCERT Ed. 2025-26, 12th, Page No. 203)

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46. (2)

Using Henry's law in the form $p = K_H x$, for the same external pressure p, the mole fraction (and hence solubility) x is inversely proportional to K_H. Since $K_H(B)$ is the smallest, gas B is the most soluble. Between A and C, $K_H(C) = 40$ kbar is smaller than $K_H(A) = 160$ kbar, so C is more soluble than A. Therefore, the order is B > C > A.

(NCERT Ed. 2025-26 12th Part-I Page No. 07)

47. (1)

The number of atoms depends on the number of moles. Option A is 3 moles directly. Option B has moles $= \frac{6}{4} = 1.5$ mol. Option D at STP has moles $= \frac{11.2}{22.4} = 0.5$ mol. Option C corresponds to mass 8u, and since one He atom has mass 4u, this is the mass of only 2 atoms, an extremely tiny fraction of a mole. Therefore, 3 mol contains the highest number of helium atoms.

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

48. (2)

A primary alkyl bromide reacts with CN^- via S_N2 to form a nitrile with one extra carbon, so 1 is pentanenitrile. Partial hydrolysis of a nitrile gives the corresponding amide, so 2 is pentanamide. Treatment of an amide with Br_2/NaOH gives Hofmann bromamide degradation, producing a primary amine with one carbon less than the amide; thus pentanamide yields butylamine. Hence the major product 3 is butylamine.

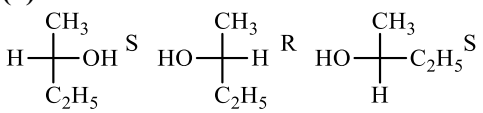
(NCERT Ed. 2025-26 12th Part-II Page No. 270)

49. (3)

In hydrogen-like (single-electron) systems, energy depends only on n, so orbitals like 3s and 3p are degenerate, making statement 2 correct and statement 3 incorrect because it claims a difference in their energies in hydrogen. The uncertainty relation in 1 is the standard form, and de Broglie's relation in 4 is also correct for a non-relativistic particle. Therefore, the wrong statement is 3.

(NCERT Ed. 2025-26 11th Part-I Page No. 46)

50. (4)



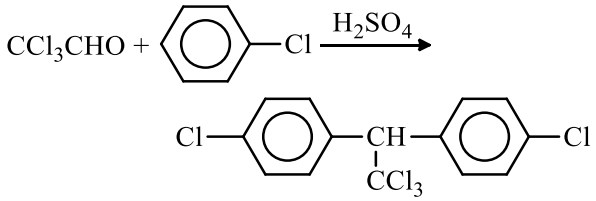
(NCERT Ed. 2025-26 12th Part-II Page No. 175)

51. (2)

Osmotic pressure is $\pi = iCRT$, and since all are non-electrolytes, $i = 1$. At the same temperature, n depends only on molarity, which for equal volume depends on moles of solute. For the same mass 9 g, moles are inversely proportional to molar mass: urea has the smallest molar mass (≈ 60), glucose is larger (≈ 180), and sucrose is the largest (≈ 342). Hence moles (and π) follow urea > glucose > sucrose, i.e., $P_2 > P_1 > P_3$.

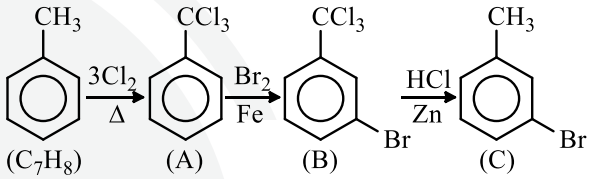
(NCERT Ed. 2025-26 12th Part-II Page No. 21)

52. (1)



(NCERT Ed. 2025-26 12th Part-II Page No. 188)

53. (3)



(NCERT Ed. 2025-26 12th Part-II Page No. 166)

54. (2)

Photoelectric emission occurs when photon energy $E = \frac{hc}{\lambda}$ is at least the work function. For λ = 500 nm, $E \approx \frac{1240}{500}\,eV = 2.48\,eV$. Metals with $W_0 \le 2.48\,eV$ will emit electrons. From the table, Li (2.30) and K (2.10) satisfy this, while Na (2.50) is slightly higher and will not. Hence, the number of metals is 2.

(NCERT Ed. 2025-26 11th Part-I Page No. 41)

55. (1)

Adding an electron is not always exothermic because some atoms (like noble gases) have positive electron gain enthalpy, so statement A is false. Removing an electron from an isolated gaseous atom requires energy, so B is always true. Due to 3p electron in Al, its first ionization energy is lower than Mg, so C is true. Electronegativity is an intrinsic property of the element (Pauling value) and does not change from one compound to another, so D is true. Electropositivity increases down group 1, so K is not the most electropositive, making E false. Therefore, the correct set is B, C and D.

(NCERT Ed. 2025-26 11th Part-I Page No. 89)

[Contd...

56. (3)

Moles of HNO_3 present are $0.50 \times 0.040 = 0.020$ mol. From the balanced equation, 2 moles of acid react with 1 mole of $CaCO_3$, so required moles of $CaCO_3$ are $0.020/2 = 0.010$ mol. Taking molar mass of $CaCO_3$ as 100 g mol^{-1}, mass of pure $CaCO_3$ needed is $0.010 \times 100 = 1.00$ g. Since the sample is 80% pure, required mass is $1.00/0.80 = 1.25$ g.

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

57. (4)

In ethane, the eclipsed conformation has maximum torsional strain due to repulsion between bonding electron pairs of the C–H bonds on adjacent carbons, making it higher in energy. The staggered conformation spreads these bonds apart as much as possible, minimizing torsional strain and stabilizing the molecule.

(NCERT Ed. 2025-26 11th Part-II Page No. 305)

58. (3)

By definition $E_{cell} = E^{red}_{cathode} - E^{red}_{anode}$. Since oxidation potential is the negative of reduction potential for the same electrode, $E^{ox}_{anode} = -E^{red}_{anode}$ so $E_{cell} = E^{red}_{cathode} + E^{ox}_{anode}$, making (B) correct. Also $E^{ox}_{cathode} = -E^{red}_{cathode}$ so $E^{ox}_{anode} - E^{ox}_{cathode} = E^{ox}_{anode} + E^{red}_{cathode} = E_{cell}$, making (D) correct. Hence the correct pair is (B) and (D).

(NCERT Ed. 2025-26 12th Part-I Page No. 33)

59. (3)

With ΔH position and ΔS negative, $\Delta G = \Delta H - T\Delta S$ any temperature non-spontaneity.

(NCERT Ed. 2025-26 11th Part-I Page No. 160)

60. (2)

Reducing sugars reduce Fehling/Tollens and all monosaccharides qualify; sucrose is nonreducing due to acetal linkage, while maltose retains a free anomeric carbon.

(NCERT Ed. 2025-26 12th Part-II Page No. 282)

61. (3)

Electrophiles are electron-pair acceptors; they may be neutral or positively charged, and they form covalent bonds by accepting from nucleophiles.

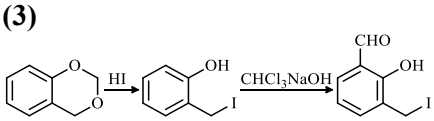
(NCERT Ed. 2025-26 11th Part-I Page No. 272)

62. (1)

Isolated systems exchange neither matter nor energy; hence no heat enters or leaves and no work is done, so $q = w = 0$.

(NCERT Ed. 2025-26 11th Part-I Page No. 138)

63. (3)



(NCERT Ed. 2025-26 12th Part-II Page No. 215)

64. (2)

For weak acid-weak base salt, $pH = 7 + \frac{pK_a - pK_b}{2} = 7 + \frac{4.6 - 5.4}{2} = 6.6$ at 25°C.

(NCERT Ed. 2025-26 12th Part-I Page No. 195)

65. (4)

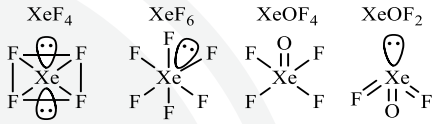
Clausius-Clapeyron gives $\frac{d(\ln P)}{dT} = \frac{\Delta H_{vap}}{RT^2}$ hence vapour pressure increases with temperature, assuming AH_{vap} constant over range, at saturation, pure substance.

66. (4)

Work is a path function; even at constant temperature, reversible and irreversible routes differ, so isothermal work isn't state function.

(NCERT Ed. 2025-26 11th Part-I Page No. 164)

67. (2)



(NCERT Ed. 2025-26 11th Part-I Page No. 124)

68. (4)

Greater metal → CO backbonding weakens C–O and lengthens it; highest electron density is in $[Fe(CO)_4]^{2-}$ so it shows maximum bond length and $[Mn(CO)_6]^+$ shows shortest

(NCERT Ed. 2025-26 12th Part-I Page No. 125)

69. (1)

Hydrogen bonding is highly directional due to orbital alignment; it's weaker than covalent O–H and needs lone pairs always present.

(NCERT Ed. 2025-26 12th Part-I Page No. 131)

70. (4)

A catalyst provides an alternative pathway with lower activation energy, leaving ΔH, ΔS and internal energy unchanged for the reaction.

(NCERT Ed. 2025-26 12th Part-I Page No. 82)

71. (3)

First order: $t = \frac{1}{k}\ln\frac{a}{a_0}$; 20 g to 5 g is fourfold, so

$t = \frac{\ln 4}{k} = \frac{0.693}{0.02} = 34.65\ s$

$t = 2 \times t_{1/2} = 69.35$

(NCERT Ed. 2025-26 12th Part-I Page No. 69)

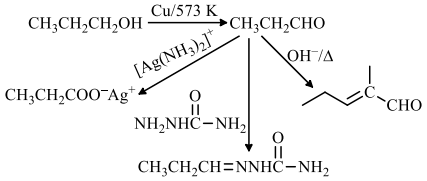
72. (2)

Cisplatin is cis-$[PtCl_2(NH_3)_2]$; the cis geometry allows DNA crosslinking, whereas trans analogs are far less active as an anticancer drug.

(NCERT Ed. 2025-26 12th Part-I Page No. 137)

[Contd...

73. (2)



(NCERT Ed. 2025-26 12th Part-II Page No. 238)

74. (1)

NaOBr oxidizes acetaldehyde to formate type products, and $LiAlH_4$ reduces carboxylate to alcohol, giving methanol.

(NCERT Ed. 2025-26 12th Part-II Page No. 240)

75. (3)

π-acceptor ligands like CO are nonclassical; calling them classical is wrong, while other statements reflect ligand-field trends in general chemistry.

(NCERT Ed. 2025-26 12th Part-I Page No. 136)

76. (1)

$Cu^{2+} \rightarrow Cu^{+}$ is favoured by stable d^{10} so IE2 is high, Mn^{2+} d^5 makes IE3 high, and Mn^{3+} readily reduces to Mn^{2+} giving positive E°; statement C is incorrect.

(NCERT Ed. 2025-26 12th Part-I Page No. 100)

77. (3)

In flame tests, Ca^{2+} emits brick-red light, whereas Na^{+} is yellow, K^{+} lilac, and Ba^{2+} apple green.

78. (3)

$Cu^{+} \rightarrow Cu^{2+} + Cu^{0}$

in (3) option Cu shows disproportionation.

(NCERT Ed. 2025-26 11th Part-II Page No. 244)

79. (1)

CH_4 is symmetrical so $\mu = 0$; in NH_3 bond moments add with lone pair, while in NF_3 they partially cancel. Hence A has maximum dipole moment

(NCERT Ed. 2025-26 12th Part-I Page No. 111)

80. (1)

Solution: Fe^{3+} forms the thiocyanato complex $Fe(SCN)^{2+}$ in solution, producing the characteristic intense blood-red colour.

81. (4)

$SnCl_4$ is molecular covalent due to high polarization and Fajan's rules; the other statements match stability and hydrolysis trends.

(NCERT Ed. 2025-26 12th Part-I Page No. 112)

82. (2)

Ambidentate ligands can bind through two different atoms. NO_2^- (N- or O-bonded), NCS^- (N-bonded isothiocyanato or S-bonded thiocyanato), CN^- (C- or N-bonded), are ambidentate. NH_3 and H_2O are not; $C_2O_4^{2-}$ is bidentate but not ambidentate; NO_3^- is typically O-donor only. Hence 4.

(NCERT Ed. 2025-26 12th Part-I Page No. 121)

83. (1)

The carius method is a high-temperature sealed-tube oxidation used mainly to estimate sulphur or halogens, not nitrogen. Option 2, 3 and 4 are correct descriptions of their respective procedures, so 1 alone is false.

(NCERT Ed. 2025-26 11th Part-II Page No. 289)

84. (2)

B → Substitution reaction to form alcohol

C → Williamson's Ether synthesis for ether

(NCERT Ed. 2025-26, 12th Part-II, Page No. 215)

85. (3)

No. of stereocentre = 3

$\therefore$ stereoisomers = $2^3 = 8$

(NCERT Ed. 2025-26 12th Part-II Page No. 177)

86. (4)

Adsorption of compound ∝ Polarity

Order by polarity a > b > c

Spend time in column a > b > c

Elution rate c > b > a.

(NCERT Ed. 2025-26 12th Part-II Page No. 279)

87. (2)

CH_3NH_2



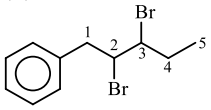
(NCERT Ed. 2025-26, 12th Part-II, Page No. 271)

88. (1)

Vitamin A	Xerophthalmia
Vitamin C	Scurvy
Vitamin D	Rickets
Vitamin K	Increased blood clotting time

(NCERT Ed. 2025-26, 12th Part-II, Page No. 296)

89. (3)



2,3-Dibromo-1-phenylpentane

(NCERT Ed. 2025-26 11th Part-II Page No. 266)

90. (3)

Higher the standard reduction potential, more the ability to get reduced and higher is the oxidising power.

(NCERT Ed. 2025-26 11th Part-I Page No. 251)

[Contd...

91. (2)
The water splitting complex is associated with the PS II, which itself is physically located on the inner side of the membrane of the thylakoid.
(NCERT Ed. 2025-26, 11th, Page No. 139)

92. (2)
The order Polymoniales includes the family Solanaceae, to which genera such as *Petunia*, *Solanum*, and *Datura* belong.
(NCERT Ed. 2022-23, 11th, Page No. 8)

93. (1)

Some frog species	Camouflage
Monarch butterfly	Poisonous/Distasteful
Cactus	Thorns
Calotropis	Cardiac glycosides

(NCERT Ed. 2025-26, 12th, Page No. 198)

94. (4)
The middle lamella is a layer mainly of calcium pectate which holds or glues the different neighbouring plant cells together.
(NCERT Ed. 2025-26, 11th, Page No. 94)

95. (3)
The sex determination in honey bee is based on the number of sets of chromosomes an individual receives. An offspring formed from the union of a sperm and an egg develops as a female (queen or worker), and an unfertilised egg develops as a male (drone) by means of parthenogenesis. This means that the males have half the number of chromosomes than that of a female. The females are diploid having 32 chromosomes and males are haploid, i.e., having 16 chromosomes. This is called as haplodiploid sex-determination system.
(NCERT Ed. 2025-26, 12th, Page No. 71)

96. (2)
A chromatographic separation of the leaf pigments shows that the colour that we see in leaves is not due to a single pigment but due to four pigments: Chlorophyll *a* (bright or blue green in the chromatogram), chlorophyll *b* (yellow green), xanthophylls (yellow) and carotenoids (yellow to yellow-orange). The chief pigment of photosynthesis is chlorophyll *a.*
(NCERT Ed. 2025-26, 11th, Page No. 137)

97. (1)
DNA as an acidic substance present in nucleus was first identified by Friedrich Meischer in 1869. He named it as 'Nuclein'.
(NCERT Ed. 2025-26, 12th, Page No. 81)

98. (2)
Michelia → carpels free (apocarpous)
Papaver → carpels fused (syncarpous)
Rose → carpels free (apocarpous)
Lotus → carpels free (apocarpous)
(NCERT Ed. 2025-26, 12th, Page No. 9)

99. (2)
Presently, 12 per cent of all bird species, 23 per cent of all mammal species, 32 per cent of all amphibian species and 31 per cent of all gymnosperm species in the world face the threat of extinction.
(NCERT Ed. 2025-26, 12th, Page No. 222)

100. (3)
In most higher plants, the growing apical bud inhibits the growth of the lateral (axillary) buds, a phenomenon called apical dominance. Removal of shoot tips (decapitation) usually results in the growth of lateral buds. It is widely applied in tea plantations and hedge-making. Auxin promote apical dominance while cytokinin is used to overcome it.
(NCERT Ed. 2025-26, 11th, Page No. 175)

101. (1)
The red thalli of most of the red algae are multicellular. Some of them have complex body organisation. The food is stored as floridean starch which is very similar to amylopectin and glycogen in structure.
(NCERT Ed. 2025-26, 11th, Page No. 27)

102. (2)
There are many traits which are not so distinct in their occurrence and are spread across a gradient. For example, in humans we don't just have tall or short people as two distinct alternatives but a whole range of possible heights. Such traits are generally controlled by three or more genes and are thus called as polygenic traits. Besides the involvement of multiple genes polygenic inheritance also takes into account the influence of environment. Human skin colour is another classic example for this.
ABO blood group is controlled by a single gene with multiple alleles.
Phenylketonuria is controlled by a single gene.
Flower colour in snapdragon is controlled by a single gene.
(NCERT Ed. 2025-26, 12th, Page No. 69)

103. (3)
Meiosis involves two sequential cycles of nuclear and cell division called meiosis I and meiosis II but only a single cycle of DNA replication.
(NCERT Ed. 2025-26, 11th, Page No. 125)

[Contd...

104. (1)

Methanobacterium is a methanogen and does not fix nitrogen.

(NCERT Ed. 2025-26, 12th, Page No. 158)

105. (4)

When fats act as respiratory substrates, more O_2 is required compared to the CO_2 produced. Thus, Respiratory Quotient (RQ) < 1.

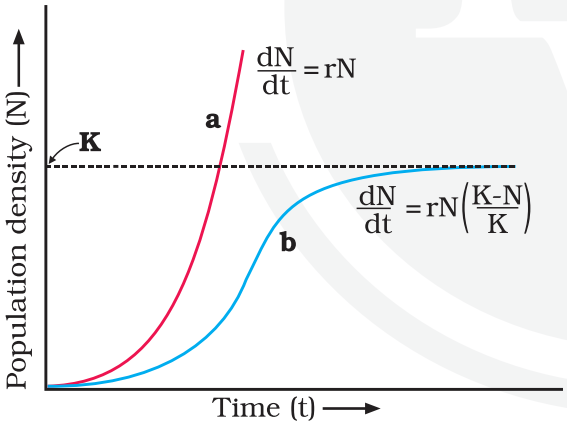
(NCERT Ed. 2025-26, 11th, Page No. 164)

106. (4)

There is a characteristic distribution of the cells within the embryo sac. Three cells are grouped together at the micropylar end and constitute the egg apparatus. The egg apparatus, in turn, consists of two synergids and one egg cell. The synergids have special cellular thickenings at the micropylar tip called filiform apparatus, which play an important role in guiding the pollen tubes into the synergid. Three cells are at the chalazal end and are called the antipodals.

(NCERT Ed. 2025-26, 12th, Page No. 11)

107. (3)



Population growth curve **a** when responses are not limiting the growth, plot is exponential, **b** when responses are limiting the growth, plot is logistic, **K** is carrying capacity

(NCERT Ed. 2025-26, 12th, Page No. 195)

108. (3)

In mango and coconut, the fruit is known as a drupe. They develop from monocarpellary superior ovaries and are one seeded. In mango the pericarp is well differentiated into an outer thin epicarp, a middle fleshy edible mesocarp and an inner stony hard endocarp. In coconut which is also a drupe, the mesocarp is fibrous.

(NCERT Ed. 2025-26, 11th, Page No. 65)

109. (3)

In a transverse section, a typical microsporangium appears near circular in outline. It is generally surrounded by four wall layers – the epidermis, endothecium, middle layers and the tapetum. The outer three wall layers perform the function of protection and help in dehiscence of anther to release the pollen. The innermost wall layer is the tapetum. It nourishes the developing pollen grains.

(NCERT Ed. 2025-26, 12th, Page No. 5)

110. (2)

The ribosome also acts as a catalyst (23S rRNA in bacteria is the enzyme- ribozyme) for the formation of peptide bond.

(NCERT Ed. 2025-26, 12th, Page No. 99)

111. (2)

The monocot stem has a sclerenchymatous hypodermis, a large number of scattered vascular bundles, each surrounded by a sclerenchymatous bundle sheath, and a large, conspicuous parenchymatous ground tissue. Vascular bundles are conjoint and closed. Peripheral vascular bundles are generally smaller than the centrally located ones. The phloem parenchyma is absent, and water-containing cavities are present within the vascular bundles.

(NCERT Ed. 2025-26, 11th, Page No. 76)

112. (3)

Scutellum	Lateral cotyledon in grasses
Coleoptile	Sheath enclosing the shoot apex
Coleorrhiza	Sheath enclosing the radicle
Epiblast	Rudimentary cotyledon

(NCERT Ed. 2025-26, 12th, Page No. 19)

113. (2)

The earliest systems of classification used only gross superficial morphological characters such as habit, colour, number and shape of leaves, etc. They were based mainly on vegetative characters or on the androecium structure (system given by Linnaeus). Such systems were artificial.

(NCERT Ed. 2025-26, 11th, Page No. 23)

114. (1)

Baculoviruses are pathogens that attack insects and other arthropods.

(NCERT Ed. 2025-26, 12th, Page No. 157)

[Contd...

115. (2)

Within the chloroplast there is membranous system consisting of grana, the stroma lamellae, and the matrix stroma. There is a clear division of labour within the chloroplast. The membrane system is responsible for trapping the light energy and also for the synthesis of ATP and NADPH. In stroma, enzymatic reactions synthesise sugar, which in turn forms starch. The former set of reactions, since they are directly light driven are called light reactions (photochemical reactions). The latter are not directly light driven but are dependent on the products of light reactions (ATP and NADPH). Hence, to distinguish the latter they are called, by convention, as dark reactions (carbon reactions). However, this should not be construed to mean that they occur in darkness or that they are not light-dependent.

(NCERT Ed. 2025-26, 11th, Page No. 136)

116. (1)

N-glycosidic linkage	Nitrogenous base to sugar
Phosphodiester bond	Between two nucleotides
Peptide bond	Between amino acids
Hydrogen bond	Between nitrogenous bases of opposite strands of DNA

(NCERT Ed. 2025-26, 12th, Page No. 80)

117. (1)

In Leguminosae, leaves are alternate, pinnately compound or sometimes simple, with a pulvinate leaf base, stipules present, and reticulate venation.

(NCERT Ed. 2022-23, 11th, Page No. 78)

118. (3)

The recent illegal introduction of the African catfish *Clarias gariepinus* for aquaculture purposes is posing a threat to the indigenous catfishes in our rivers.

(NCERT Ed. 2025-26, 12th, Page No. 223)

119. (2)

Elaioplasts	Store oils and fats
Amyloplasts	Store carbohydrates
Aleuroplasts	Store proteins
Chromoplasts	Contain fat-soluble carotenoid pigments

(NCERT Ed. 2025-26, 11th, Page No. 97)

120. (1)

Since genes are known to be located on chromosomes, alteration in chromosomes results in abnormalities or aberrations. Chromosomal aberrations are commonly observed in cancer cells.

(NCERT Ed. 2025-26, 12th, Page No. 72)

121. (1)

In ascomycetes (e.g., *Claviceps*), karyogamy is followed by meiosis, leading to the formation of haploid ascospores produced endogenously inside an ascus.

Claviceps and *Neurospora*→ Ascomycete → Endogenous spores (ascospores)

Ustilago → Basidiomycete → Exogenous spores

Puccinia → Basidiomycete → Exogenous spores

Alternaria → Deuteromycete → No known sexual stage

(NCERT Ed. 2025-26, 11th, Page No. 18)

122. (2)

The living differentiated cells, that have lost the capacity to divide can regain the capacity of division under certain conditions. This phenomenon is termed as dedifferentiation. For example, formation of meristems – interfascicular cambium and cork cambium from fully differentiated parenchyma cells.

(NCERT Ed. 2025-26, 11th, Page No. 172)

123. (3)

Streptokinase produced by the bacterium *Streptococcus* and modified by genetic engineering is used as a 'clot buster' for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack. Another bioactive molecule, cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus *Trichoderma polysporum*. Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents.

(NCERT Ed. 2025-26, 12th, Page No. 153)

124. (2)

Mesophyll in dorsiventral leaf has two types of cells – the palisade parenchyma and the spongy parenchyma. The adaxially placed palisade parenchyma is made up of elongated cells, which are arranged vertically and parallel to each other. The oval or round and loosely arranged spongy parenchyma is situated below the palisade cells and extends to the lower epidermis.

(NCERT Ed. 2025-26, 11th, Page No. 76)

[Contd...

125. (3)

Orchid on a mango branch → Commensalism
Barnacles on a whale's back → Commensalism
Cuscuta growing on a hedge plant → Parasitism
Venus flytrap and insect → Predation

(NCERT Ed. 2025-26, 12th, Page No. 200)

126. (3)

In oocytes of some vertebrates, diplotene can last for months or years.

(NCERT Ed. 2025-26, 11th, Page No. 126)

127. (4)

The split-gene arrangements represent probably an ancient feature of the genome. The presence of introns is reminiscent of antiquity, and the process of splicing represents the dominance of RNA-world. In eukaryotes, the monocistronic structural genes have interrupted coding sequences – the genes in eukaryotes are split. The coding sequences or expressed sequences are defined as exons. Exons are said to be those sequence that appear in mature or processed RNA. The exons are interrupted by introns. Introns or intervening sequences do not appear in mature or processed RNA.

(NCERT Ed. 2025-26, 12th, Page No. 93)

128. (3)

The TCA cycle starts with the condensation of acetyl group with oxaloacetic acid (OAA) and water to yield citric acid. The reaction is catalysed by the enzyme citrate synthase and a molecule of CoA is released.

(NCERT Ed. 2025-26, 11th, Page No. 158)

129. (4)

Both bryophytes and pteridophytes share several features:
Water is required for fertilisation (motile antherozoids)
Multicellular sporophyte is present in both groups
Archegonia are present in both
However, the key difference:
In bryophytes, the sporophyte is dependent on the gametophyte.
In pteridophytes, the sporophyte is independent.

(NCERT Ed. 2025-26, 11th, Page No. 29)

130. (4)

Micro-organisms such as *Lactobacillus* and others commonly called lactic acid bacteria (LAB) grow in milk and convert it to curd. During growth, the LAB produce acids (during lactic acid fermentation) that coagulate and partially digest the milk proteins. A small amount of curd added to the fresh milk as inoculum or starter contain millions of LAB, which at suitable temperatures multiply, thus converting milk to curd, which also improves its nutritional quality by increasing vitamin B_{12}. In our stomach too, the LAB play very beneficial role in checking disease-causing microbes.

(NCERT Ed. 2025-26, 12th, Page No. 151)

131. (1)

Chromosome 1	Chromosome with the highest number of genes
Chromosome 11	Location of genes for β-thalassemia
Chromosome 16	Location of gene for α-thalassemia
Chromosome 21	Chromosome involved in Down's syndrome

(NCERT Ed. 2025-26, 12th, Page No. 75)

132. (1)

Primary productivity depends on the plant species inhabiting a particular area. It also depends on a variety of environmental factors, availability of nutrients and photosynthetic capacity of plants. Therefore, it varies in different types of ecosystems.

(NCERT Ed. 2025-26, 12th, Page No. 207)

133. (2)

Very often, red dinoflagellates (Example: *Gonyaulax*) undergo such rapid multiplication that they make the sea appear red (red tides).

(NCERT Ed. 2025-26, 11th, Page No. 15)

134. (2)

Some amino acids are coded by more than one codon, hence the code is degenerate.

(NCERT Ed. 2025-26, 12th, Page No. 96)

135. (3)

GA_3 is used to speed up the malting process in brewing industry.

(NCERT Ed. 2025-26, 11th, Page No. 176)

[Contd...

136. (3)

Members of phylum Aschelminthes (now commonly referred to as Nematoda) have a cylindrical body that appears circular in cross-section, which is why they are called roundworms. Roundworms are bilaterally symmetrical and triploblastic, but they are not true coelomates. They possess a pseudocoelom (false body cavity). Hence, they are classified as pseudocoelomates, not true coelomates.

(NCERT Ed. 2025-26, 11th, Page No. 43)

137. (2)

Squamous epithelium consists of thin, flat cells with irregular boundaries, forming a delicate lining (e.g., blood vessels).

Simple cuboidal epithelium consists of one layer of cube-shaped cells, commonly found in kidney tubules.

PCT cells possess microvilli forming a brush border to increase absorption.

Ciliated epithelium occurs mainly in respiratory tract and fallopian tubes, not generally on the inner surface of all hollow organs.

Compound epithelium mainly provides protection, not secretion/absorption.

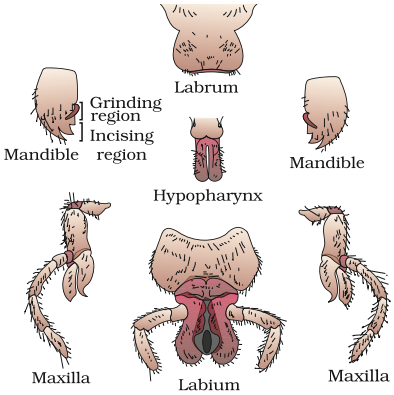
(NCERT Ed. 2020-21, 11th, Page No. 101)

138. (3)

In Phylum Porifera (sponges), water enters through minute pores (ostia) into a central cavity (spongocoel) and goes out through the osculum. This water transport or canal system is helpful in food gathering, respiratory exchange, and removal of waste. Sponges have a cellular level of organization and lack a nervous system to generate nerve impulses.

(NCERT Ed. 2025-26, 11th, Page No. 40)

139. (2)



The mouthparts of a cockroach consist of a labrum (upper lip), a pair of mandibles, a pair of maxillae, and a labium (lower lip). A median flexible lobe acting as a tongue is called the hypopharynx, which lies within the cavity enclosed by the mouthparts.

(NCERT Ed. 2020-21, 11th, Page No. 112)

140. (1)

List-I	List-II
Coelenterata	Cnidoblasts
Ctenophora	Comb plates
Platyhelminthes	Flat body, suckers
Echinodermata	Water vascular system

(NCERT Ed. 2025-26, 11th, Page No. 51)

141. (3)

Gap junctions facilitate cell-to-cell communication by connecting the cytoplasm of adjoining cells, allowing rapid transfer of ions, small molecules, and sometimes big molecules. Tight junctions prevent leakage, and adhering junctions perform cementing functions.

(NCERT Ed. 2020-21, 11th, Page No. 102)

142. (1)

Thyroid has follicles filled with colloid (for thyroxine storage) and stromal connective tissue.

Exophthalmic goitre (Graves' disease) is caused by hyperthyroidism, not hypothyroidism.

PTH increases blood calcium levels by acting on bones, kidneys, and intestines.

Thyrocalcitonin (calcitonin) is a peptide hormone, not steroidal.

Cortisol regulates metabolism, stress response, and catabolic processes.

(NCERT Ed. 2025-26, 11th, Page No. 245)

143. (3)

ELISA is a serological test that detects the presence of antigens or antibodies in a sample using the specific binding between an antigen and its antibody.

An enzyme attached to the antibody or antigen produces a color change, indicating a positive reaction.

PCR: Amplifies DNA sequences; it does not rely on antigen-antibody interaction.

Autoradiography: Uses radioactive probes to detect nucleic acids or proteins.

Gel Electrophoresis: Separates molecules based on size/charge, not antigen-antibody binding.

(NCERT Ed. 2025-26, 12th, Page No. 183)

[Contd...

144. (3)

List-I	List-II
Alkaloids	Morphine, Codeine
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Polymeric substances	Rubber, gums, cellulose

(NCERT Ed. 2025-26, 11th, Page No. 108)

145. (3)

A sparged stirred-tank bioreactor allows sterile air bubbles to pass through the culture, which increases the oxygen transfer area, ensuring efficient oxygen supply for aerobic microbial or cell cultures.

Downstream processing is not limited to separation. It includes a series of steps: separation, purification, concentration, and sometimes formulation of the product to obtain the final pure product.

(NCERT Ed. 2025-26, 12th, Page No. 174)

146. (2)

In Diabetes mellitus, due to insulin deficiency or resistance:

- Glucose appears in urine (glycosuria) because blood glucose exceeds the renal threshold.
- Ketone bodies appear (ketonuria) due to excessive fat breakdown for energy in the absence of adequate glucose utilization.

(NCERT Ed. 2025-26, 11th, Page No. 213)

147. (4)

By the end of about 24 weeks (end of second trimester), the body is covered with fine hair, eye-lids separate, and eyelashes are formed. By the end of nine months of pregnancy, the foetus is fully developed and is ready for delivery.

(NCERT Ed. 2025-26, 12th, Page No. 38)

148. (2)

- Human ovaries typically release 1 egg per month (ovulation).
- Identical (monozygotic) twins arise from a single fertilized egg splitting, so still only 1 egg is released.
- Fraternal (dizygotic) twins develop from two separate eggs, so 2 eggs are released and fertilized.

(NCERT Ed. 2025-26, 12th, Page No. 33)

149. (2)

During chemical transformation, bacterial cells (usually *E. coli*) are made competent using $CaCl_2$.

A brief heat shock at 42°C creates a thermal imbalance across the membrane, allowing the plasmid DNA to enter the cell.

Temperatures like 72°C or 94°C would kill the cells, and 37°C is too mild to induce DNA uptake.

(NCERT Ed. 2025-26, 12th, Page No. 171)

150. (1)

The theory of spontaneous generation, which claimed that life can arise from non-living matter, was eventually disproved.

Louis Pasteur performed experiments using pre-sterilized swan-neck flasks. He showed that microorganisms appeared only when pre-existing life (microbes) could enter the flask, proving that life arises from existing life, not spontaneously.

(NCERT Ed. 2025-26, 12th, Page No. 111)

151. (3)

Metastasis is a hallmark of malignant tumors. Tumor cells detach from the primary site, enter the blood or lymph, travel to distant organs, and form secondary tumors.

(NCERT Ed. 2025-26, 12th, Page No. 141)

152. (2)

In sliding filament theory, actin (thin) filaments slide over myosin (thick) filaments.

Thick filaments do not change in length, so the A band stays the same.

I band and H zone shorten as the filaments slide past each other, bringing Z lines closer.

This results in muscle shortening without shortening the filaments themselves.

(NCERT Ed. 2025-26, 11th, Page No. 222)

153. (1)

120 mm Hg is systolic, not diastolic; hypertension is >130/80 mm Hg.

Atherosclerosis deposits plaques in arteries, narrowing the lumen.

Angina can occur in adults of any age, not only the elderly.

Heart failure is when the heart cannot pump blood effectively.

Heart attack is due to blocked blood flow, not the heart stopping; cardiac arrest is when it stops.

(NCERT Ed. 2025-26, 11th, Page No. 203)

[Contd...

154. (3)

The majority of CO_2 is transported as bicarbonate ions.

- In tissues, CO_2 reacts with water to form carbonic acid (H_2CO_3), catalyzed by carbonic anhydrase.
- Carbonic acid dissociates into H^+ and HCO_3^- (bicarbonate), which diffuses into plasma.
- Only small amounts are transported dissolved in plasma or bound to hemoglobin.

(NCERT Ed. 2025-26, 11th, Page No. 189)

155. (2)

The Bidder's canal in frogs functions as a passage for sperms. It connects the vasa efferentia of the testes to the urino-genital duct, running through the kidney. This allows sperm to travel from the testes to the cloaca for external fertilization.

(NCERT Ed. 2025-26, 11th, Page No. 84)

156. (2)

The cerebral cortex is grey matter, consisting mainly of neuron cell bodies; white matter lies beneath, made of myelinated axons.

The limbic system and hypothalamus regulate emotions, motivation, and sexual behaviour, integrating hormonal and neural signals.

The midbrain has four rounded swellings, called corpora quadrigemina (2 superior colliculi for vision, 2 inferior colliculi for hearing).

The cerebellum surface is highly convoluted, forming folia, which increase surface area for neurons and coordination of movement.

The medulla oblongata is continuous with the spinal cord, transmitting sensory and motor signals between brain and body.

(NCERT Ed. 2025-26, 11th, Page No. 236)

157. (3)

The pineal gland is located on the dorsal side of the forebrain. It secretes melatonin, which plays a crucial role in regulating the 24-hour diurnal rhythms of the body, including the sleep-wake cycle and body temperature.

(NCERT Ed. 2025-26, 11th, Page No. 242)

158. (1)

Passive immunity occurs when ready-made antibodies are given, e.g., antivenom or immunoglobulin therapy; the body doesn't produce them. Thermostable DNA polymerase (*Taq* polymerase) survives the high temperatures of PCR denaturation, allowing repeated cycles without enzyme inactivation.

(NCERT Ed. 2025-26, 12th, Page No. 136)

159. (3)

IC, EC, and VC can be measured by a spirometer, as they involve air that can be inhaled or exhaled voluntarily. FRC (Functional Residual Capacity) includes residual volume, the air always remaining in lungs, which a spirometer cannot measure directly.

(NCERT Ed. 2025-26, 11th, Page No. 187)

160. (2)

Humulin is human insulin produced by recombinant DNA technology. It is identical to natural human insulin, so it does not trigger unwanted immune responses.

Maintaining constant body temperature is a feature of homeotherms, humans and aves.

(NCERT Ed. 2025-26, 12th, Page No. 182)

161. (3)

HIV specifically infects $CD4^+$ helper T-cells, gradually reducing their number. This leads to weakened immunity and susceptibility to opportunistic infections, causing AIDS.

(NCERT Ed. 2025-26, 12th, Page No. 140)

162. (4)

In blue-white screening, if the gene is inserted into β-galactosidase, the enzyme is disrupted, and colonies appear white, not blue.

White muscle fibres (fast-twitch) have few mitochondria and rely on anaerobic respiration.

(NCERT Ed. 2025-26, 11th, Page No. 223)

163. (2)

Parasympathetic nerves (mainly via the vagus nerve) release acetylcholine, which slows the sinoatrial (SA) node, reducing heart rate. It does not significantly increase cardiac output or stroke volume.

(NCERT Ed. 2025-26, 11th, Page No. 202)

164. (4)

List-I (Application of transgenic animals)	List-II (Specific example/description)
Normal physiology and development	Studying complex factors like insulin-like growth factor
Study of disease	Serving as models for Alzheimer's and Cystic fibrosis
Biological products	Producing human protein alpha-1-antitrypsin for emphysema
Vaccine safety	Using mice to test batches of the polio vaccine

(NCERT Ed. 2025-26, 12th, Page No. 184)

[Contd...

165. (2)

Vaults are cervical caps inserted into the female reproductive tract to cover the cervix during coitus. They act as a barrier contraceptive, preventing sperm from entering the uterus.

(NCERT Ed. 2025-26, 12th, Page No. 44)

166. (3)

Enzymes do not change the energy of substrates or products. They speed up reactions by lowering the activation energy, making it easier for the reaction to occur.

(NCERT Ed. 2025-26, 11th, Page No. 115)

167. (4)

Tetany occurs when blood calcium (Ca^{2+}) levels drop sharply, leading to involuntary muscle spasms and cramps. It is distinct from myasthenia gravis or muscular dystrophy, which have different causes.

(NCERT Ed. 2025-26, 11th, Page No. 227)

168. (3)

In host-delivered RNA interference (RNAi), plants are engineered with nematode-specific genes. These genes produce both sense and antisense RNA, which hybridize to form double-stranded RNA (dsRNA). When the nematode ingests dsRNA, it triggers gene silencing, reducing nematode survival and infestation. This method is specific and non-toxic to the plant and other organisms.

(NCERT Ed. 2025-26, 12th, Page No. 180)

169. (2)

At higher altitudes, oxygen concentration in the air is lower. To maintain oxygen supply to tissues, the respiratory rate and depth increase (hyperventilation). This helps meet the body's oxygen demand despite reduced atmospheric oxygen.

(NCERT Ed. 2025-26, 11th, Page No. 190)

170. (3)

Ball and socket joint: Humerus and pectoral girdle.
Hinge joint: Knee joint
Pivot joint: Allows rotation, e.g., between atlas and axis, not carpals
Saddle joint: Between carpal and metacarpal of thumb

(NCERT Ed. 2025-26, 11th, Page No. 227)

171. (1)

List-I	List-II
Gastrin	Secreted by GIT
Amygdala	Part of limbic system
ZIFT	ART
Myometrium	Middle layer of uterus

(NCERT Ed. 2025-26, 12th, Page No. 30)

172. (2)

Male gametes (sperm) are formed in the seminiferous tubules, not in the interstitial spaces. Leydig cells in the interstitial spaces secrete androgens (testosterone), which stimulate spermatogenesis in the seminiferous tubules.

(NCERT Ed. 2025-26, 12th, Page No. 31)

173. (1)

In Miller's experiment, a reducing atmosphere with methane (CH_4), hydrogen (H_2), ammonia (NH_3), and water vapour was maintained. The temperature ~800°C simulated early Earth conditions, allowing the formation of organic compounds like amino acids.

(NCERT Ed. 2025-26, 12th, Page No. 111)

174. (4)

Inulin is a fructose polymer stored in plants (like chicory, dahlia, and onion) as an energy reserve. Starch is a glucose polymer, glycogen is stored in animals, and cellulose provides structural support.

(NCERT Ed. 2025-26, 11th, Page No. 110)

175. (3)

List-I	List-II
Sporozoites	Infective stage for humans
Pons	Consists of fibre tracts
Glucagon	Hyperglycemic
Aldosterone	Mineralocorticoid

(NCERT Ed. 2025-26, 12th, Page No. 101)

176. (3)

Genital herpes is caused by herpes simplex virus (HSV), which remains latent in nerve cells.
Even if detected early, it cannot be completely cured, though antiviral drugs can control outbreaks. Other STIs like syphilis, chlamydiasis, and gonorrhoea can be fully cured with antibiotics if treated promptly.

(NCERT Ed. 2025-26, 12th, Page No. 136)

177. (3)

Let p = frequency of dominant allele, q = frequency of recessive allele.
Given q = 0.4, so $p = 1 - q = 1 - 0.4 = 0.6$.
Frequency of heterozygotes (2pq) = $2 \times 0.6 \times 0.4 = 0.48$.

(NCERT Ed. 2025-26, 12th, Page No. 121)

[Contd...

178. (3)

Collagen is the most abundant protein in the animal world, whereas RuBisCO is the most abundant protein in the whole of the biosphere.

(NCERT Ed. 2025-26, 11th, Page No. 110)

179. (1)

Homologous structures → same origin/ancestry, may perform different functions. Both thorns and tendrils are modified shoots, so they are homologous. Both are anatomically similar and diverged for different functions (defense vs climbing).

(NCERT Ed. 2025-26, 12th, Page No. 115)

180. (2)

Aschelminthes have separate sexes (dioecious). A key feature of sexual dimorphism in this group is that females are often longer than males.

(NCERT Ed. 2025-26, 11th, Page No. 43)

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